MAKSAT SATYBALDIEV

HIGH ENERGY ASTROPHYSICIST

maksat.satybaldiev@ntnu.no
https://maksatsat.github.io

EXPERIENCE

January 2026	**Visiting Researcher** | University of Geneva, ISDC, Versoix, Switzerland
2023 *to present*	**PhD Candidate** | Norwegian University of Science and Technology, Department of Physics
2020 *to* 2023	**Research Assistant** | Space Research Institute of the Russian Academy of Sciences (IKI), Laboratory of Fundamental and Applied X-ray Astrophysics

EDUCATION

2023 *to present*	**PhD in Astrophysics** **Norwegian University of Science and Technology**, Trondheim, Norway Supervisors: Prof. Manuel Linares and Dr. Vittoria Vecchiotti
2021 *to* 2023	**MS Applied Mathematics and Physics** **Moscow Institute of Physics and Technology**, Russia Thesis: *"Study of X-ray flares from wind-fed X-ray binaries"* Supervisor: Dr. Ilya Mereminskiy
2017 *to* 2021	**BS Applied Mathematics and Physics** **Moscow Institute of Physics and Technology**, Russia Thesis: *"Periodic activity of X-ray bursters"* Supervisor: Dr. Ilya Mereminskiy

PUBLICATIONS

1. **Satybaldiev, M.** & Linares, M., *"Gamma-Ray Modes in a Transitional Pulsar"*, **The Astrophysical Journal Letters** 1011, L57 (2026)
2. Due, B., Bridziute, J., Linares, M., Turchetta, M. & **Satybaldiev, M.**, *"Optical Follow-Up of Two Spider Pulsar Candidates: An Irradiated Redback and a Possible Double Giant Impostor"*, **The Astrophysical Journal** (2026)
3. **Satybaldiev, M.**, Linares, M., & Vecchiotti, V., *"Gamma-Ray Orbital Modulation in Spider Pulsars: Three Discoveries and a Universal Modulated Fraction"*, **The Astrophysical Journal**, 998, 94 (2026)
4. **Satybaldiev, M. N.**, Mereminskiy, I. A., Lutovinov, A. A., Karasev, D. I., Semena, A. N., & Shtykovsky, A. E., *"Spectral and Temporal Analysis of the Supergiant Fast X-ray Transient IGR J16195-4945 with SRG/ART-XC,"* **Astronomy Letters**, 49, 249 (2023)

AWARDS & SCHOLARSHIPS

November 2025	**Best young scientist of the year** (second place), Ministry of Science, Higher Education, and Innovation of the Kyrgyz Republic

April 2023	**Best presentation award winner**, 65th MIPT Conference
Spring 2023	**Scholarship** for excellent academic achievements, MIPT
2017 *to* 2023	**State Academic Scholarship** for bachelor and master students at MIPT

Conferences, Workshops, Seminars & Extra Courses

August 2026	**Neutron Stars and Nebulae**, Skukuza, Kruger National Park, South Africa *"Gamma-Ray Modes in a Transitional Pulsars"*(solicited talk)
January 2026	**University of Geneva**, Versoix, Switzerland *"Gamma-Ray Orbital Modulation in Spider Pulsars"*(seminar)
September 2025	**Pulsar 2025**, Geremeas, Italy *"Gamma-Ray Orbital Modulation in Spider Pulsars: Results from a Systematic Search"* (poster)
June 2025	**European Astronomical Society Annual Meeting**, Cork, Ireland *"Discovery of gamma-ray orbital modulation in three spider pulsars"* (contributed talk)
June 2025	**Norwegian Physical Society Meeting**, Trondheim, Norway *"Discovery of gamma-ray orbital modulation in three spider pulsars"* (contributed talk)
May 2025	**XVIII Neutron Star Workshop**, Bonn, Germany *"Discovery of gamma-ray orbital modulation in three spider pulsars"* (contributed talk)
June 2024	**IPTA 2024**, Milan, Italy International Pulsar Timing Array Student Week (participant)
November 2023	**NTNU Astro & Theory Seminar**, Trondheim, Norway *"Study of X-ray flares from the wind-fed X-ray binaries"* (seminar)
July 2023	**Physics of Neutron Stars**, St. Petersburg, Russia *"Spectral and Temporal Analysis of the Supergiant Fast X-ray Transient IGR J16195-4945 with SRG/ART-XC"* (poster)
April 2023	**XX Young Scientists Conference**, Space Research Institute (IKI) *"Flares from the wind-fed X-ray binaries with red and blue supergiants"* (contributed talk)
April 2023	**65th MIPT All-Russian Scientific Conference in honor of the 115th anniversary of L.D. Landau**, Moscow Institute of Physics and Technology *"Flares from the wind-fed X-ray binaries with red and blue supergiants"* (contributed talk)
December 2022	**High Energy Astrophysics Today and Tomorrow**, Space Research Institute (IKI) *"Spectral and temporal analysis of the SRG/ART-XC observation of the Supergiant Fast X-ray Transient IGR J16195-4945"* (poster)

November 2022	**High Energy Astrophysics Department Seminar**, Space Research Institute (IKI) *"Spectral and timing analysis of the SRG/ART-XC observation of the Supergiant Fast X-ray Transient IGR J16195-4945"* (contributed talk)
June 2022	**Code/Astro Workshop**, Caltech Software Engineering Workshop for Astronomy (participant)
April 2022	**XIX Young Scientists Conference**, Space Research Institute (IKI) *"SRG/ART-XC observation and analysis of the X-ray transient IGR J16195-4945"* **M. N. Satybaldiev**, I. A. Mereminskiy (contributed talk)
April 2021	**XVIII Young Scientists Conference**, Space Research Institute (IKI) *"Periodic activity of the X-ray bursters"* (contributed talk)
August 2020	**Data-driven Astronomy**, Coursera Python, SQL, Machine Learning applied to Astronomy

Academic Service

2026 *to present*	**RAS Techniques and Instruments (RASTI) referee**
2025 *to present*	**NewAthena Science Community member**
2025 *to present*	**Astro & Theory Seminars** and **NTNU Astroparticle Journal Club** organizer Organizing seminars and the journal club at the Astrophysics and Particle Physics Group, NTNU
2024 *to present*	**European Astronomical Society (EAS) member**

Technical Experience

Telescope Experience	***Fermi-LAT, XMM-Newton, NICER, NuSTAR, SRG/ART-XC, Swift/BAT, Swift/XRT, INTEGRAL, RXTE*** Timing and spectral analysis.
Computing skills	**Programming**: Python, C **Utilities**: astropy, git, numpy, XSPEC, DS9 **Presentation**: LaTeX, MatPlotLib, Adobe Photoshop **OS**: macOS, Linux, Windows

Teaching

Fall 2022	**Teaching Assistant**, Space Physics Department Introductory Course Moscow Institute of Physics and Technology
January 2020	**Intensive Course in Olympiad-level Physics for High School Students** School-Lyceum #61, Bishkek, Kyrgyzstan

Other Information

Biographical Information

Citizenship: Kyrgyz Republic (Kyrgyzstan)
Languages: Kyrgyz, English, Russian, Norwegian (basic), French (basic)
Birth: 30 September 1999, Bishkek, Kyrgyzstan

Social Activities

Scientific Olympiad Organizer
Organization of scientific olympiads for high school students (2017–2021)

Election Observer
2020 Kyrgyz parliamentary election

This CV was last updated on November 11, 2026.